CMPS4191 | UNIVERSITY OF BELIZE

Advanced Web Technologies

Short Polling vs. Long Polling

An Empirical Evaluation of HTTP Observation Mechanics, Trade-offs, & Protocol Selection

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ARCHITECTURAL MECHANICS

Implementation Differences

Short Polling

Stateless Timer Loop

Client Loop: Timer-driven checks (at 1,000 ms) `setInterval`

Server Behavior: Immediate DB query; instantly returns regardless of state change. `200 OK`

Lifecycle: Fixed sleep interval between request cycles.

Long Polling

Event-Driven Connection Hold

Client Loop: Event-driven; re-issues request immediately upon response receipt.

Server Behavior: Connection held on Go channel until DB commit or 20s timeout (). `204 No Content`

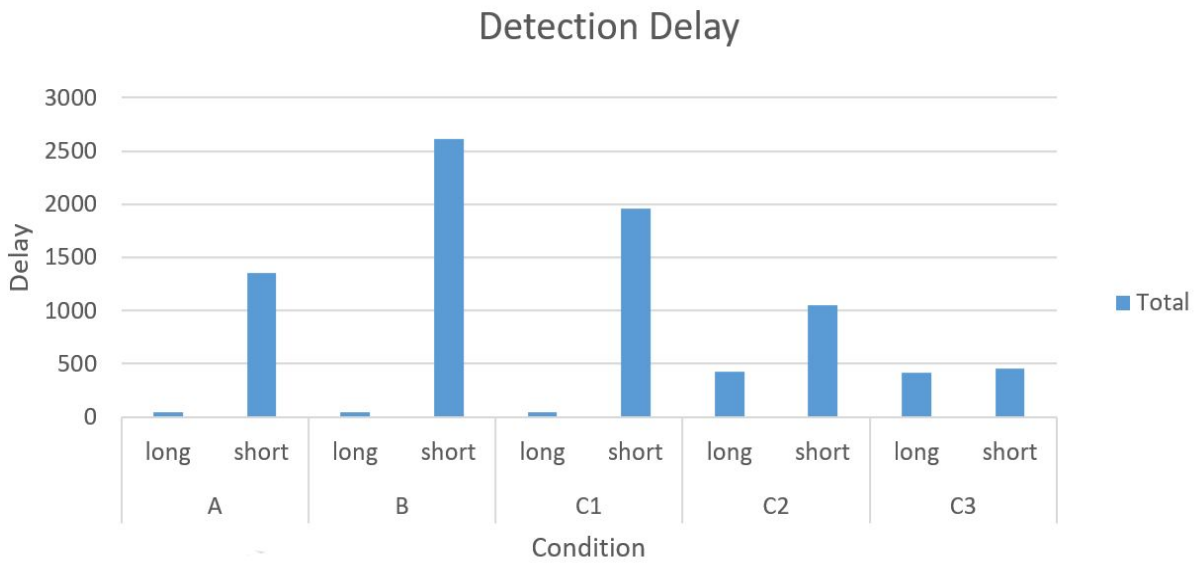
Lifecycle: Eliminates timer delay between backend event and browser notification.

EMPIRICAL BENCHMARKS

Completion-Detection Delay

Long Polling (14 ms) — 98% Faster

Short Polling (846 ms)



LONG POLLING LATENCY

14 ms

Near-instant socket flush on commit.

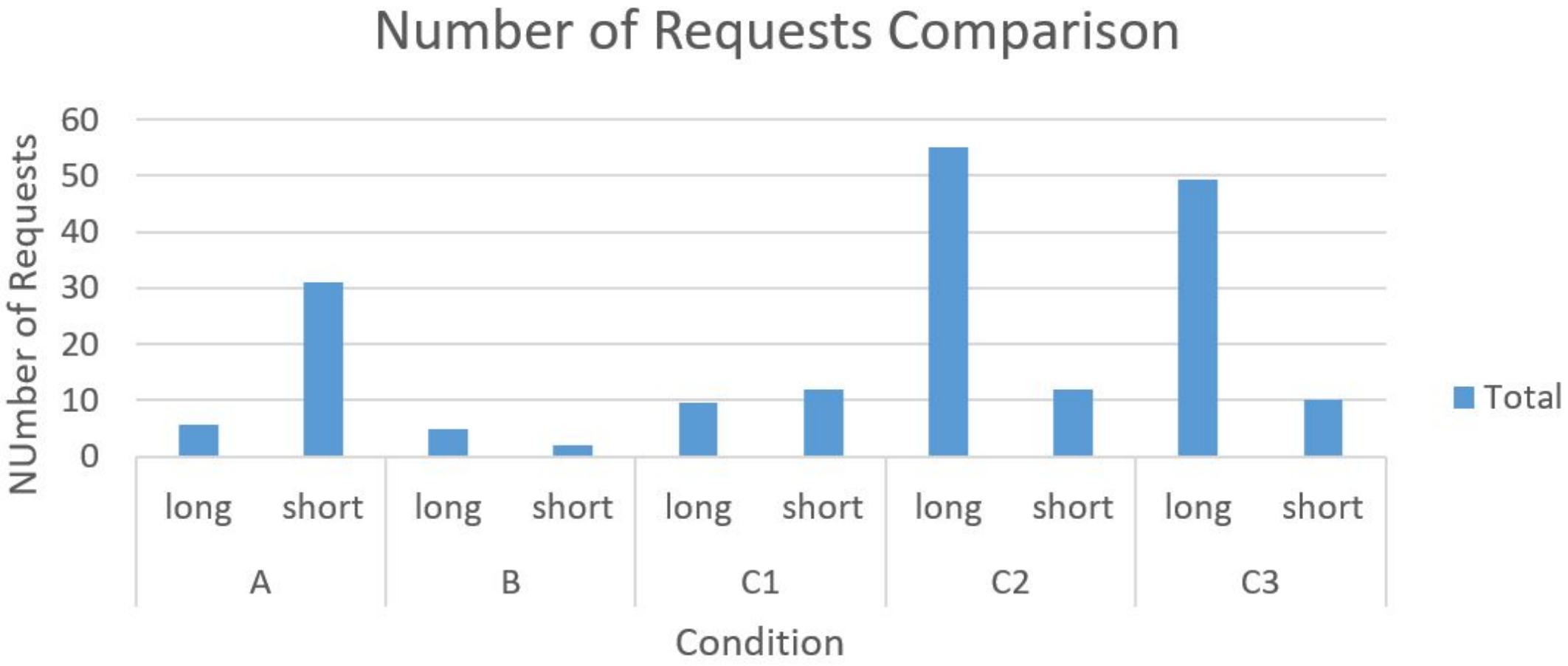
SHORT POLLING LATENCY

846 ms

Timer phase delay bounded by 1,000 ms.

TIME-SCALING ANALYSIS

Job Duration vs. Total Requests

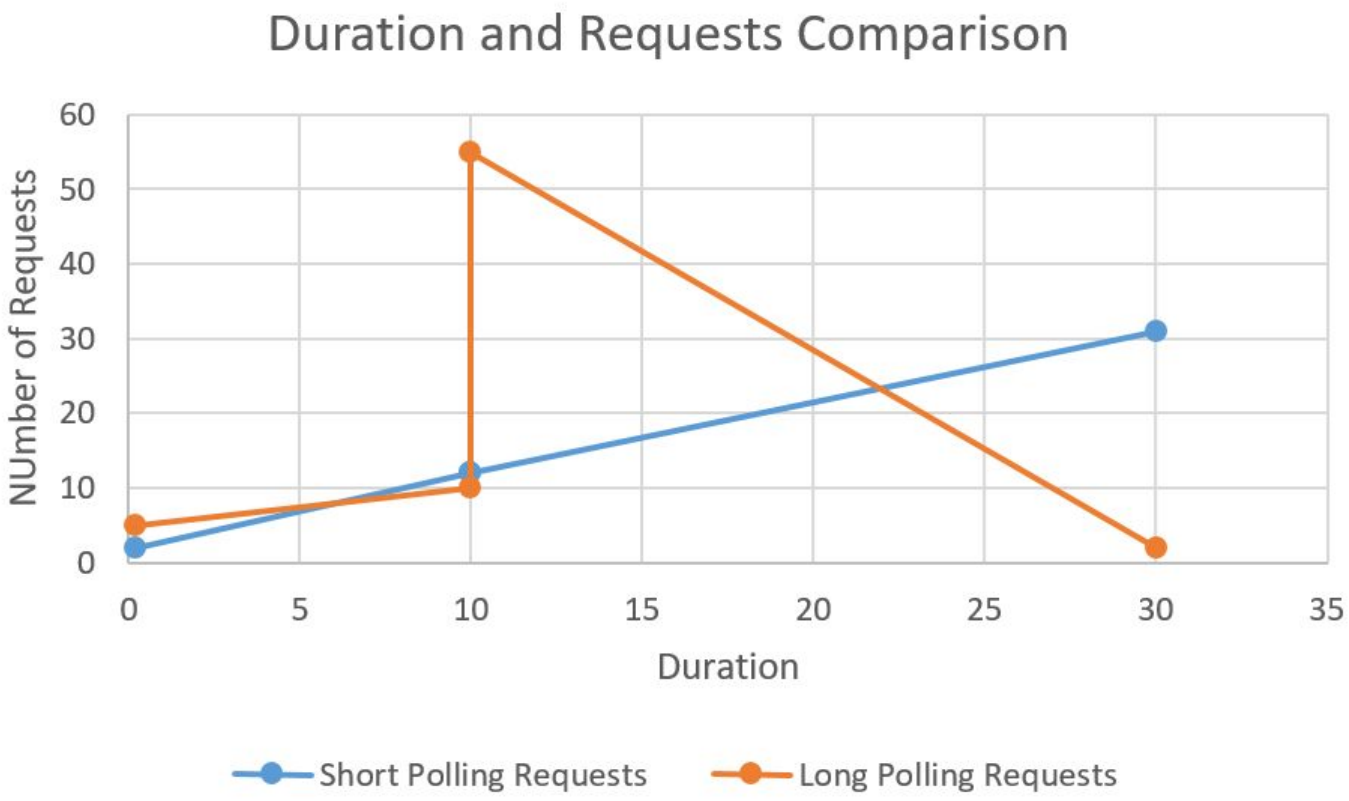


Total requests vary, why?

TIME-SCALING ANALYSIS

Job Duration vs. Total Requests

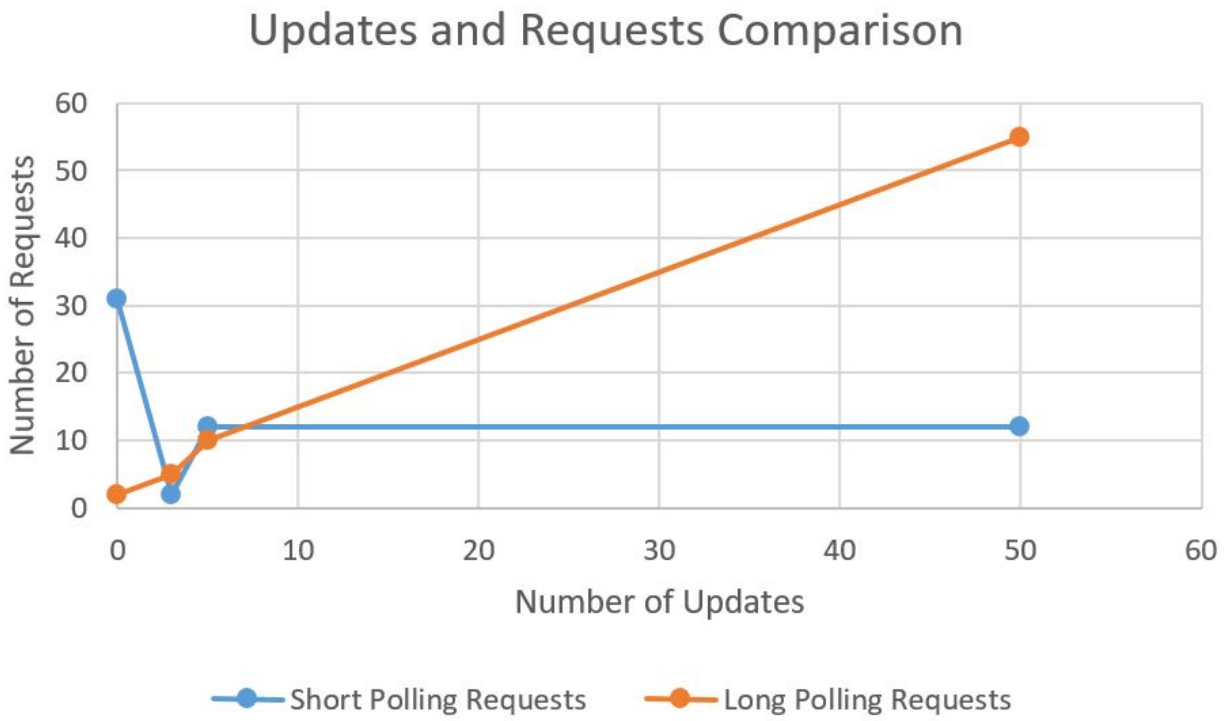
Short Polling scales with job duration. Long Polling does not



EVENT-SCALING ANALYSIS

Backend Updates vs. Total Requests

Long Polling scales with STATE UPDATES. Short polling does not.



PROTOCOL ASSESSMENT

Long Polling: Advantages & Disadvantages

Advantages

~14 ms Detection Awareness: Delivers terminal state changes near-instantly upon database commit.

Zero Idle Traffic: Eliminates redundant polling checks during quiet processing windows.

100% Version Coverage: Captures every intermediate progress milestone under low latency.

Disadvantages

Request Thrashing: High update frequencies force rapid request churn (55 reqs / 10s).

Server Resource Footprint: Held HTTP connections consume open server sockets and goroutines.

Middlebox Risk: Network proxies and firewalls may drop idle held connections unexpectedly.

ARCHITECTURAL DECISION GUIDE

When to Use Which Protocol?

🕓 Choose Short Polling When...

High Update Frequency: State changes occur rapidly (>1 update/sec) where long polling would thrash.

Non-Time-Critical Latency: Delayed completion detection is acceptable (e.g., daily batch reports).

Massive Scale & Socket Limits: Serving high concurrency with limited server socket capacity.

⚡ Choose Long Polling When...

Instant Completion Required: Real-time awareness is needed without WebSocket / SSE overhead.

Variable Idle Durations: Jobs experience unpredictable processing times (e.g., image jobs, AI generation).

Bandwidth Conservation: Eliminating idle polling traffic is prioritized.

200 OK

IT’S DEMO TIME!